

DSA 8420 Spring 2025

Project 3

Due April 29, 2025

We have explored in class how to solve a standard Sudoku puzzle using Pyomo by formulating it as an integer programming problem. In this project, you will take on a creative extension of that idea: solving a Killer Sudoku puzzle using Pyomo.

Task Begin by reading the introduction to Killer Sudoku at <https://sudoku.com/killer>. (If you have a moment, try playing a game or two to get a feel for the puzzle!) Your task is to write a Pyomo model to solve Killer Sudoku puzzles. Your implementation should be general and user-friendly — designed so that someone with no background in optimization or Pyomo can follow your instructions and modify the data in your Jupyter Notebook to solve *any* instance of a Killer Sudoku puzzle.

Submission. Submit your completed Jupyter Notebook as a file named `sumdoku.ipynb` **via Canvas**. In the notebook (or in a separate accompanying document), include clear instructions explaining how users can adapt your code to input and solve an *arbitrary* Killer Sudoku puzzle.

Hints.

1. Consider using binary variables to represent each possible digit in a cell. Specifically, define binary variables x_{ijk} to indicate whether cell (i, j) contains the digit k . You can also define integer variables $y_{ij} \in \{1, \dots, 9\}$ for each cell, where $y_{ij} = k$ if and only if $x_{ijk} = 1$. The

relationship between x and y can then be expressed as:

$$y_{ij} = \sum_{k=1}^9 k x_{ijk}.$$

2. Use the following Killer Sudoku puzzle to test your code:

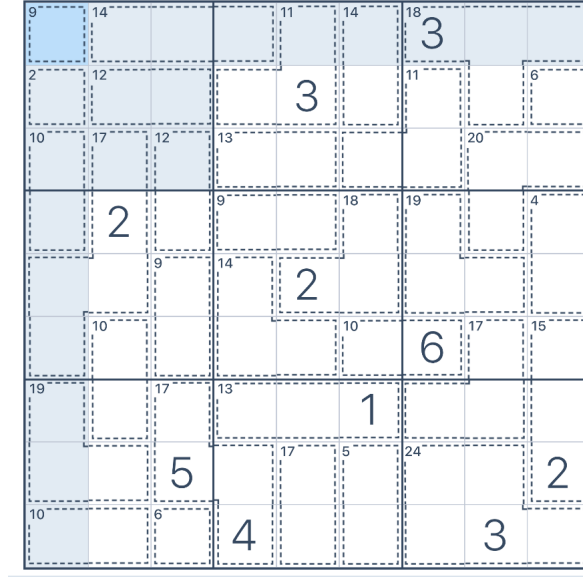


Figure 1: From: <https://sudoku.com>

Below is an example format for representing the input data. Feel free to adjust the structure as needed to best suit your implementation.

known = [(1, 7, 3), (2, 5, 3), (4, 2, 2), (5, 5, 2), (6, 7, 6), (7, 6, 1), (8, 3, 5), (8, 9, 2), (9, 4, 4), (9, 8, 3)]

cage = [[(1,1)], [(1,2), (1,3), (1,4)], [(1,5), (2,4), (2,5)], [(1,6), (2,6)], [(1,7), (1,8), (1,9),(2,8)], [(2,1)], [(2,2), (2,3)], [(2,7), (3,6), (3,7)], [(2,9)], [(3,1), (4,1)], [(3,2), (4,2), (5,1), (5,2), (6,1)], [(3,3), (4,3)], [(3,4), (3,5)], [(3,8), (3,9), (4,8)], [(4,4), (4,5)], [(4,6), (5,5), (5,6)], [(4,7), (5,7), (5,8)], [(4,9), (5,9)], [(5,3), (6,3)], [(5,4), (6,4), (6,5)], [(6,2), (7,2)], [(6,6), (6,7)], [(6,8), (7,7), (7,8)], [(6,9), (7,9), (8,9)], [(7,1), (8,1), (8,2)], [(7,3), (8,3), (8,4), (9,4)], [(7,4), (7,5), (7,6)], [(8,5), (9,5)], [(8,6), (9,6)], [(8,7), (8,8), (9,7), (9,8), (9,9)], [(9,1), (9,2)], [(9,3)]]

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cage_sum = [9, 14, 11, 14, 18, 2, 12, 11, 6, 10, 17, 12, 13, 20, 9, 18, 19,  
4, 9, 14, 10, 10, 17, 15, 19, 17, 13, 17, 5, 24, 10, 6]
```